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www.elsevier.com/locate/caor

PII: S0305-0548(16)30042-9
DOI: http://dx.doi.org/10.1016/j.cor.2016.02.018
Reference: CAOR3948

To appear in: *Computers and Operation Research*

Received date: 12 May 2015
Revised date: 18 January 2016
Accepted date: 26 February 2016

Cite this article as: Belma Turan, Stefan Minner and Richard F. Hartl, A VNS approach to multi-location inventory redistribution with vehicle routing *Computers and Operation Research,* http://dx.doi.org/10.1016/j.cor.2016.02.018

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A VNS approach to multi-location inventory redistribution with vehicle routing

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Abstract

This paper analyzes a two-echelon inventory problem with uncertain demand, where perishable products are distributed from a central warehouse to retail stores. To better deal with uncertainty, we propose a model in which, at a certain point in time during the operation time window, transshipments between retail stores are allowed in order to re-balance the inventories. The model captures the following decisions: (i) which retail stores are included in re-balancing, (ii) in which sequence will they be visited, (iii) what are the optimal initial delivery and transshipment quantities and (iv) what is the optimal timing for transshipments. The model is solved with variable neighborhood search, which is extended with a dynamic programming sub-routine to handle inventory allocation. We compare our results with similar models from literature. The results show a noticeable improvement over the standard "single-order, no transshipments" policy even though only the inventory level of the visited customer is known. The transshipment quantities have to be determined without knowledge of the inventories of all other customers.

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1 Introduction

The distribution of perishable products from a central warehouse to retail stores is a complex process as demand for products is uncertain and the quality of products worsens with time. In a typical supply chain, all decisions about delivery quantities to each retail store have to be made a-priori, i.e. before any demand information is revealed. As the leftover inventory usually has no value on the next day and any unsatisfied demand is lost, deciding about the right delivery quantity is a very challenging task.

Several models that help companies to better deal with uncertainty have been proposed. In our previous work, Turan et al. (2015), we considered an option for delivering an additional resupply quantity to selected retail stores during the day. The work was inspired by a real-world problem concerning the delivery of bakery products and can be applied to a variety of other problems where arrival of new fresh products might be a trigger for additional sales.

Another option that allows more flexibility in decision making is including inventory transshipments, as they help with the re-balancing of inventory levels at a pool of retail stores by allowing them to collaborate and use the same inventory. Transshipments might also be useful for perishable seasonal products or weekly specials. However, transshipments cannot be used for perishable items with an inherent risk of being contaminated during additional packaging and transportation, e.g. bread and bakery products. Allowing transshipments between the retail stores leads to a reduction in overall inventory and lead times, as the stores do not again have to be replenished by the central warehouse, and to an increase in overall profit due to the risk-pooling effect. However, when including transshipments, we should not neglect the resulting transportation costs, since the transshipments only pay off if the additional benefit is higher than the additional costs. Therefore, we need an integrated approach that will simultaneously optimize the routing and the inventory allocation problem.

We propose a routing problem with transshipments that includes several decisions: (i) how to select the pool of retail stores for inventory re-balancing, (ii) how to find the right sequence in which these stores will be visited and determine the timing for re-balancing, (iii) how to find

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the best initial supply quantities for each retail store, and (iv) how to determine the optimal transshipment quantity for each retail store on every route for a fixed route and timing of visit. Note that transshipment quantities have to be determined without knowledge of the inventories of other retail stores left to be visited on the same route. We capture these decisions along with imposed restrictions in a mathematical model with the goal of maximizing the entire expected profit, i.e. net revenues minus procurement and transportation costs.

To solve the mathematical model, we develop a solution method that is time-efficient but still provides high-quality solutions. For the routing decisions, (i) and (ii), we chose variable neighborhood search (VNS), which was proposed by Mladenović and Hansen (1997). The VNS is easy to understand and use, and it has already been efficiently applied to various routing problems (see Hansen and Mladenović (2001) or Hansen et al. (2010)). The inventory allocation decisions (iii) and (iv) are optimized using stochastic dynamic programming.

The remainder of the paper is organized as follows: In Section 2, we discuss the literature related to our problem. We present the mathematical formulation of the problem in Section 3. The solution method is introduced in Section 4, followed by numerical results in Section 5 and the conclusion and discussion about future work in Section 6.

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2 Review of Related Literature

Allowing transshipments between retailers in order to efficiently manage inventories and reduce costs has already been intensively studied in the literature. Some of the first contributions in this field are by Allen (1958), Gross (1963) and Das (1975), who find the optimal policy that minimizes total costs for problems with two retail stores. The basic model has been extended in many ways: By considering the multiple-retailers problem (see Karmarkar and Patel (1977)), the multi-period problem (see Karmarkar (1981)), or even by discussing a decentralized system (Lee and Whang (2002)). A recent review of existing literature is given in Paterson et al. (2011). They classify the literature according to eleven characteristics.

Based on the type of transshipments, we distinguish between proactive and reactive transship-

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ments. While reactive transshipments are considered after demand has been realized, proactive transshipments are used to re-balance stock levels at some point in time and are therefore suitable for our problem. The timing of the transshipment plays an important role in the performance of the selected transshipment policy, but the optimal timing is hard to determine. Some of the studies assume that transshipments take place at the beginning of the period (see Allen (1958), Allen (1962)), while others select a fixed point in time somewhere within the period (see Agrawal et al. (2004)). Agrawal et al. (2004) also consider the optimization of the transshipment time by dividing the periods into sub-periods and solving the lost sales minimization problem using dynamic programming. Kiesmüller and Minner (2009) compare the fixed timing of transshipments determined a-priori with a flexible transshipment policy with demand update and show that flexibility in decision making is beneficial, especially when demand is highly uncertain and profit margins are low.

The majority of the models studied in the literature assume that the transshipment transportation times can be neglected. However, there are some contributions that show how important transportation times and costs might be. Jönsson and Silver (1987) assume the same transportation time for all locations and find the optimal balance between additional costs and the target service level. Bertrand and Bookbinder (1998) consider a cost-minimization problem with non-identical transportation costs. They use a simulation model to solve the problem and determine the service level. Tagaras and Vlachos (2002) deal with non-identical transportation times and additionally address the importance of type and variability of the assumed demand distribution. Nonås and Jörnsten (2005) present a greedy heuristic to determine the optimal transshipment policy. This heuristic even provides optimal solutions for a case with two or three retail stores. They extend this work in Nonås and Jörnsten (2007), presenting the optimal policies for four retail stores and discussing a setting for an optimal policy in case of n locations. Smirnov and Gerchak (2014) study the performance of circular unidirectional chaining in case of symmetric transshipment costs. They state that, in the case of non identical demand distribution, one should first solve the sequencing problem at the network design stage. In Lien et al. (2011), the authors discuss

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transshipment network design and highlight the difficulty of finding the right visiting sequence, as the optimal traveling salesman solution is not necessarily the best solution for transshipment networks.

Our problem belongs to the field of inventory routing problems with stochastic demand. Coelho et al. (2013) give a comprehensive review of the literature on inventory routing problems. Different inventory allocation models for problems with perishable products are presented in Turan (2016).

Coelho et al. (2012) are the first to formulate the inventory routing problem with transshipments. They adapt the standard inventory routing problem in such a way that transshipments are allowed and propose an adaptive large neighborhood search as a solution method. Shen et al. (2011) solve a multi-period multi-mode problem of transporting crude oil from a supply center to multiple customers by a Lagrangian relaxation approach. Recently, Mirzapour Al-e hashem and Rekik (2014) formulated a multi-product multi-period inventory routing problem with transshipments that they manage to solve to optimality for small instances.

On top of the routing and allocation decisions, we also optimize the selection of retail stores to be visited for transhipment. Hence, our problem closely relates to routing problems with profits (see Flood (1956), Cook et al. (2008), Reinelt (1991), Toth and Vigo (2014), Christofides (1976), Laporte (1992), Golden et al. (2008)). Recent reviews on the routing problem with profits are given by Feillet et al. (2005) and Archetti et al. (2013). Diverse solution methods have been applied to a variety of routing problems with profits. A fast and effective VNS is proposed by Vansteenwegen et al. (2009b), which also inspired our solution approach. VNS has also been successfully applied to inventory routing problems (see Popović et al. (2012)).

The novelty of our research lies in the combination of transshipment network design with customer selection and the simultaneous optimization of the underlying routing problem. To the best of our knowledge, a such model has not been studied before.

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3 Mathematical Formulation

We consider a two-echelon retail chain model consisting of one central warehouse that replenishes n retail stores. Each retail store i sells $\ell = 1, .., m$ perishable products in a given assortment with stochastic iid period demand. The unit sales price p_ℓ and the unit procurement costs c_ℓ are assumed to be the same for each store. Demand $d_{i\ell}^{[t_1,t_2]}$ is modeled during time interval $[t_1, t_2]$ with density function $f_{i\ell}(\bullet)^{[t_1,t_2]}$ and cumulative density function $F_{i\ell}(\bullet)^{[t_1,t_2]}$. Furthermore, we use $\mu_{i\ell}$ and $\sigma_{i\ell}$ to denote the mean value and the standard deviation of demand. We focus on daily demand at each store and therefore normalize the demand period in the time interval $[0, 1]$, where 0 denotes the opening and 1 the closing time of the store. The leftover inventory at the end of a period has no value on the next day. Furthermore, we assume that any unmet demand cannot be backordered or substituted.

Each store i receives a delivery quantity, $Q_{i\ell}^0$, of product ℓ before time 0. We define t_i as a point in time during time interval $[0, 1]$ when store i re-balances its inventories with transshipments. The transshipments of product ℓ at time t_i are denoted by $Q_{i\ell}^{t_i}$. If the initial shipment to retail store i is too low, and we expect that the inventory at time t_i is not enough to meet the demand in the remaining period $[t_i, 1]$, then transshipments will serve to add items to the stock at the store, i.e. the transshipment quantity will be positive. On the other hand, if the initial shipment to retail store i is too high, and we expect leftovers after completing demand in the remaining period $[t_i, 1]$, then transshipments will serve to reduce the stock of the store by taking away the surplus and reassigning it to other stores, i.e. the transshipment quantity will have a negative value.

Transshipments are executed by a homogenous fleet of vehicles $k = 1, .., r$. All vehicles start and end their routes at depot 0. Let t_{0k} be the dispatch time of vehicle k and $a_{i\ell k}$ the current stock of product ℓ on vehicle k upon visiting retail store i. The sequencing decisions are captured by a binary flow variable, x_{ijk}, which takes value 1 if store j is visited immediately after store i on route k. The impact of the routing is measured in two ways: With tt_{ij}, we express the transportation time and with tc_{ij}, we denote the transportation costs between retail stores i and j. With ST, we denote the service time needed to unload or load the transshipment quantity, which is assumed

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to be the same for all retail stores.

The total expected sales of product ℓ achieved through transshipments on route k are denoted by $\bar{V}_{k\ell}^{[t_{0k},1]}$ and depend on the dispatch time of vehicle, t_{0k}. The total expected sales are found after maximizing the expected sales at each retail store visited on the same route k. Let $V_{i\ell}^{[t_i,1]}(a_{i\ell k}, y_{i\ell}^{t_i}, Q_{i\ell}^{t_i}, Q_{i\ell}^{0})$ denote the expected sales through transshipments of product ℓ from retail store i onwards. The sales are dependent on the current stock on the vehicle, $a_{i\ell k}$, the initial delivery quantity, $Q_{i\ell}^{0}$, the inventory level at time t_i, $y_{i\ell}^{t_i}$, and the size of the transshipment $Q_{i\ell}^{t_i}$. Moreover, let $\dot{V}_{i\ell}^{[t_i,1]}(a_{i\ell k}, y_{i\ell}^{t_i}, Q_{i\ell}^{0})$ be the expected sales for optimal transshipment quantity at each retail store and, further, let $\ddot{V}_{i\ell}^{[t_i,1]}(a_{i\ell k}, Q_{i\ell}^{0})$ be the unconditional sales whenever we express the expected inventory level as a function of demand until time t_i, $d_{i\ell}^{[0,t_i]}$, $y_{i\ell}^{t_i} = (Q_{i\ell}^{0} - d_{i\ell}^{[0,t_i]})^{+}$.

We propose the following mathematical model:

$$\max \underbrace{\sum_{\ell=1}^{m}\sum_{i=1}^{n}\left((p_\ell - c_\ell)Q_{i\ell}^{0} - p_\ell \sum_{d_{i\ell}^{[0,t_i]}=0}^{Q_{i\ell}^{0}-1} F_{i\ell}^{[0,t_i]}(d_{i\ell}^{[0,t_i]})\right)}_{\mathbf{O}_1} + \underbrace{\sum_{\ell=1}^{m}\sum_{k=1}^{r} p_\ell \bar{V}_{k\ell}^{[t_{0k},1]}}_{\mathbf{O}_2} - \underbrace{\sum_{i=0}^{n}\sum_{j=0}^{n}\sum_{k=1}^{r} t c_{ij} x_{ijk}}_{\mathbf{O}_3} \quad (1)$$

$$\sum_{k=1}^{r}\sum_{j=1}^{n} x_{0jk} = \sum_{k=1}^{r}\sum_{i=1}^{n} x_{i0k} \leq r \quad (2)$$

$$\sum_{i=0}^{n} x_{ijk} = \sum_{i=0}^{n} x_{jik} \leq 1 \qquad \forall j = 1,..,n, \forall k = 1,..,r \quad (3)$$

$$t_i + tt_{ij} + ST \leq t_j + M(1 - x_{ijk}) \qquad \forall i = 0,..,n, \forall j = 0,..,n, \forall k = 1,..,r \quad (4)$$

$$t_i + tt_{ij} + ST \geq t_j - M(1 - x_{ijk}) \qquad \forall i = 0,..,n, \forall j = 0,..,n, \forall k = 1,..,r \quad (5)$$

$$t_{0k} + tt_{0j} \leq t_j + M(1 - x_{0jk}) \qquad \forall j = 0, .., n, \forall k = 1, .., r \tag{6}$$

$$t_{0k} + tt_{0j} \geq t_j - M(1 - x_{0jk}) \qquad \forall j = 0, .., n, \forall k = 1, .., r \tag{7}$$

$$\bar{V}_{k\ell}^{[t_{0k},1]} = \max_{0 \leq Q_{i\ell}^0} \Big\{ \sum_{i=1}^{n} \ddot{V}_{i\ell}^{[t_i,1]}(0, Q_{i\ell}^0) x_{0ik} \Big\} \qquad \forall k = 1, .., r \tag{8}$$

$$\ddot{V}_{i\ell}^{[t_i,1]}(a_{i\ell k}, Q_{i\ell}^0) = \Big(1 - F_{i\ell}^{[0,t_i]}(Q_{i\ell}^0)\Big) \dot{V}_{i\ell}^{[t_i,1]}(a_{i\ell k}, 0, Q_{i\ell}^0) +$$
$$+ \sum_{d_{i\ell}^{[0,t_i]}=0}^{Q_{i\ell}^0} f_{i\ell}^{[0,t_i]}(d_{i\ell}^{[0,t_i]}) \dot{V}_{i\ell}^{[t_i,1]}(a_{i\ell k}, Q_{i\ell}^0 - d_{i\ell}^{[0,t_i]}, Q_{i\ell}^0) \qquad \forall i = 1, .., n, \forall \ell = 1, ., .m, \forall k = 1, .., r \tag{9}$$

$$a_{i\ell k} - Q_{i\ell}^{t_i} \leq a_{j\ell k} + M(1 - x_{ijk}) \qquad \forall i = 0, .., n, \forall j = 1, .., n, \forall \ell = 1, .., m, \forall k = 1, .., r \tag{10}$$

$$a_{i\ell k} - Q_{i\ell}^{t_i} \geq a_{j\ell k} - M(1 - x_{ijk}) \qquad \forall i = 0, .., n, \forall j = 1, .., n \forall \ell = 1, ..m, \forall k = 1, .., r \tag{11}$$

$$a_{i\ell k} \leq M(1 - x_{0ik}) \qquad \forall i = 1, .., n, \forall \ell = 1, .., m \tag{12}$$

$$Q_{i\ell}^{t_i} \geq a_{i\ell k} - M(1 - x_{i0k}) \qquad \forall i = 1, .., n, \forall \ell = 1, .., m, \forall k = 1, .., r \tag{13}$$

$$\dot{V}_{i\ell}^{[t_i,1]}(a_{i\ell k}, y_{i\ell}^{t_i}, Q_{i\ell}^0) = \max_{-y_{i\ell}^{t_i} \leq Q_{i\ell}^{t_i} \leq a_{i\ell k}} V_{i\ell}^{[t_i,1]}(a_{i\ell k}, y_{i\ell}^{t_i}, Q_{i\ell}^{t_i}, Q_{i\ell}^0) \qquad \forall i = 1, .., n, \forall \ell = 1, .., m, \forall k = 1, .., r \tag{14}$$

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$$V_{i\ell}^{[t_i,1]}(a_{i\ell k}, y_{i\ell}^{t_i}, Q_{i\ell}^{t_i}, Q_{i\ell}^{0}) = Q_{i\ell}^{t_i} + y_{i\ell}^{t_i} - \sum_{d_{i\ell}^{[t_i,1]}=0}^{Q_{i\ell}^{t_i}+y_{i\ell}^{t_i}-1} F_{i\ell}^{[t_i,1]}(d_{i\ell}^{[t_i,1]}) +$$

$$+ \sum_{j=1}^{n} \ddot{V}_{j\ell}^{[t_i,1]}(a_{i\ell k} - Q_{i\ell}^{t_i}, Q_{j\ell}^{0}) x_{ijk} \qquad \forall i = 1,..,n, \forall \ell = 1,..,m, \forall k = 1,..,r \tag{15}$$

$$x_{ijk} \in \{0,1\}, \quad 0 < t_i < 1, \quad 0 < t_{0k} < 1, \qquad \forall i = 0,..,n, \forall j = 0,..,n,$$

$$Q_{i\ell}^{0}, y_{i\ell}^{t_i}, a_{i\ell k} \in \mathbb{N}_0, \quad Q_{i\ell}^{t_i} \in \mathbb{Z} \qquad \forall \ell = 1,..,m, \forall k = 1,..,r \tag{16}$$

The objective of our model (1) is to maximize the expected total profit: Revenues achieved through the initial shipment reduced by procurement costs (term $\mathbf{O}_1$), plus the additional revenues achieved through transshipments (term $\mathbf{O}_2$), minus the transportation costs for executing the transshipments (term $\mathbf{O}_3$). The costs for the initial delivery are fix and therefore omitted from consideration. Constraints (2) to (7) are needed for the routing part of the problem. The maximal number of used vehicles cannot exceed r (constraints (2)). Constraints (3) are in- and out-bound constraints and constraints (4) and (5) ensure a feasible visiting schedule. Each route starts and ends at depot 0 and no location is visited more than once. Note that it is possible not to visit a customer. The departure time from the depot for each route is calculated by constraints (6) and (7).

Constraints (8) to (15) depict the second stage of the problem: Once the sequence of the retail stores is known, we can optimize delivery quantities. During the inventory allocation, no re-sequencing is allowed. Constraints (8) calculate the expected sales on each route by maximizing the expected sales at the first retailer. With constraints (9), we calculate the expected sales at each store i for every product ℓ. Constraints (10) and (11) show how the remaining stock on the vehicle changes after visiting a retailer. The vehicle arrives at the first retail store with no stock, as shown by constraints (12). When arriving at the last retail store, all the remaining stock is assigned to this store, as in (13). The size of transshipment to all other stores is obtained by

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solving the Bellman equation, (14). Note that the size of positive transshipments is bounded by the remaining stock on the vehicle and the size of negative transshipments is bounded by the expected remaining inventory level at the retail store. The transshipment size is chosen in such a way that it maximizes the expected sales at both current store i and the pool of stores still to be visited by the same vehicle, as ensured by (15). The inventory allocation is explained in detail in Section 4.2.

In our model, we assume that vehicles have no capacity constraints. During re-balancing, only small amounts are moved around, whereas the timing of the re-balancing is critical in order to gain additional benefit. Therefore, the timing imposes a route duration constraint, which is stronger than a vehicle capacity constraint.

We further use the assumption that the first retail store should receive no re-fill. If there are no capacity limitations on inventory level and if the surplus in inventory could be reassigned to other stores without extra costs, then additional re-fill probably needed could already be delivered with the initial supply. Therefore, the initial stock on the vehicle before visiting the first retail store is equal to zero (constraints (12)).

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4 Solution Method

For solving the underlying routing problem, we chose VNS. The basic idea of VNS is the interchanging of two phases: During the **Local Search** phase, the local optimum is obtained, whereas the solution is perturbed during the **Shaking** phase. We adapt this method to suit our problem and propose Algorithm 1. The challenges of the model are the following ones: First, we select retail stores that will take part in inventory re-balancing and find the sequence in which these retailers will be visited. Further, we have to determine the timing of vehicle dispatch and the sizes for both initial shipment quantity and transshipment quantity. The standard VNS is therefore extended by additional operators, which allow us a selection of retail stores to be visited, and by an inventory allocation operator. As the objective value strongly depends on the revenues obtained by inventory allocation, we solve this subproblem to optimality.

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Algorithm 1 Variable neighborhood search

Initialization of solution s:
Regret Insertion
Allocation
$ALG_LOOP = 0$
repeat
if $ALG_LOOP = DISTURB_LOOP$ **then**
Disturb
Worst
Random
Allocation
end if
repeat
Shaking: Select random solution s_h at h^{th} neighborhood of s:
Inter-route Two-Opt
Inter-route Swap
Inter-route Relocate
Allocation
Local Search: find local optimum, s_h^*:
Intra-route Two-Opt (Allocation)
Intra-route Swap (Allocation)
Intra-route Relocate (Allocation)
if Solution is accepted **then**
$s^* = s_h^*$
$h = 1$
else
$h = h + 1$
end if
until $h = h^{max}$
repeat
Re-insert
Local search: find local optimum, s'
if *improvement* **then**
$s^* = s'$
end if
until no more *improvement*
if Solution is accepted **then**
$s = s^*$
end if
$ALG_LOOP + 1$
until $ALG_LOOP > MAX_LOOP$ or *Stopping criterion*

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4.1 Initialization

During the **Initialization** phase, we create the initial solution. The first decision is to determine the starting number of vehicles needed to execute the transshipments. Two conflicting effects regarding the number of vehicles used and the number of retailers visited by a vehicle can be observed: If more than one retail store is visited on the same route, the risk-pooling effect is higher. However, some stores might not benefit from transshipments as they are visited too early or too late. Therefore, we let the number of vehicles vary during the optimization process. As for the initial number of vehicles, r, we assume that all retail stores should be visited more or less close to the middle of the working day (operation time window). Hence, a reasonable tour length will not be longer than half the length of the operation time window. We propose the following approximation for the initial number of vehicles:

$$r = \lceil \frac{n * (tt^{avg} + ST)}{0.5} \rceil \tag{17}$$

where tt^{avg} is the average distance between all retail stores and ST is the service time needed to load/ unload the vehicle.

After obtaining the initial number of routes, r, we also select r seed customers, being the retail stores located furthest away from the depot, and assign them to the routes. Additional locations are selected by applying the **Regret Insertion** operator. The benefit of the **Regret** heuristic on vehicle routing problems has been outlined by Trick (1992) and recently by Pisinger and Ropke (2007). As opposed to the usual calculations of regret value Δ, we calculate it as the difference not only between the best and the second best route, but also as the difference between the best and the second best position in the best route. The procedure is depicted in Figure 1. Let a be the costs of inserting retailer i at the best position, A, in the best route 1. Let b be the costs for inserting the retailer at the second best position in route 1, B. Moreover, let c be the costs for inserting the retailer at the best insertion position C in the second best route (route 2). The regret value is calculated as follows:

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$$\Delta = (c - a) + (b - a) \tag{18}$$

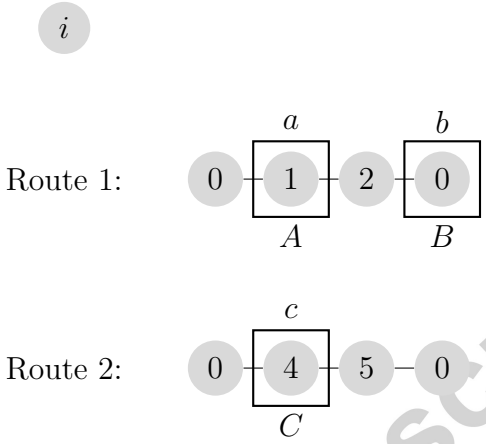


Figure 1: Regret Insertion operator

The retail store with the highest regret value is inserted into the best route at the best position and regret values have to be recalculated. The procedure stops when all stores have been inserted. Note that the best position is calculated only by using additional transportation costs that arise when this location is visited. The resulting benefit in revenues is calculated after all retailers have been inserted through the **Allocation** operator.

4.2 Allocation

The profit is measured as the total revenues minus the total procurement and transportation costs. The transportation costs result from the given retailer's sequence. The total revenues and procurement costs depend on the allocation decisions, i.e. the sizes of the initial shipments and the additional transhipment decisions.

An illustration of the problem with $n = 5$ retail stores is given in Figure 2. All stores receive an initial shipment $Q^0_{i\ell}$ of product ℓ at time 0. At time t_{0k}, vehicle k leaves the depot in order to re-balance the inventories of the stores. Note that in this example only stores 3, 4 and 5 have been selected for the transshipments and are going to be visited in that same sequence. Vehicle k

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leaves the depot empty and arrives at store 3 at time t_3. The timing of the succeeding location, t_j, can be calculated as $t_j = t_i + tt_{ij} + ST$ if location j is visited immediately after location i and tt_{ij} is the travel time between the two locations. For the first location, location 3, we have to determine the transshipment size, $Q_{3\ell}^{t_3}$. At the first location, the transshipment size has to be either negative, i.e. we have to take some items from this location, or zero. This quantity is equal to the stock on the vehicle we have upon arriving at retailer 4, $-Q_{3\ell}^{t_3} = a_{4\ell k}$. The decision about the transshipment quantity at each retailer i is made based on the observed inventory level of this retail store, $y_{i\ell}$, the probability distribution of the inventory levels of all other succeeding stores on the same tour and the remaining stock on the vehicle, $a_{i\ell k}$. When arriving at the last retailer, retailer 5, we just assign all the remaining stock to this retailer, $Q_{5\ell}^{t_3} = a_{5\ell k}$, as it becomes worthless afterwards.

Note that decisions about timing of the second delivery and initial shipment have to be made at time 0, before any demand has been realized. For a given route and fixed timing of dispatch, we have to determine the sizes of the initial delivery, only relying on the distribution information on demand. At time t_i we visit retailer i and make a decision about transshipment based on the size of the initial shipment, the observed inventory level, the current stock on the vehicle and the probability distribution of the inventory levels at the remaining retail stores on the route.

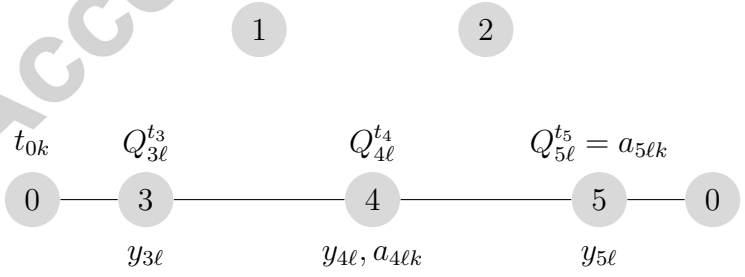


Figure 2: Inventory allocation

4.2.1 Dynamic Programming Sub-Routine

We solve the allocation problem by using stochastic dynamic programming with backward recursion. We apply the procedure presented in Figure 2 and the model presented in (8) - (15). For

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the general model, we assume that the last retail store on route k has index η_k. All the remaining stock of product ℓ upon visiting retailer η_k will be assigned to this retailer, $q_{\eta_k\ell} = a_{\eta_k\ell k}$. The expected sales for the last store in time period $[t_{\eta_k}, 1]$ are calculated as follows:

$$\begin{aligned}\dot{V}_{\eta_k\ell}^{[t_{\eta_k},1]}(a_{\eta_k\ell k}, y_{\eta_k\ell}, Q^0_{\eta_k\ell}) &= a_{\eta_k\ell k} + y_{\eta_k\ell} - \sum_{i=0}^{a_{\eta_k\ell k}+y_{\eta_k\ell}-1} \sum_{d_{\eta_k\ell}^{[t_{\eta_k},1]}=0}^{i} f_{\eta_k\ell}^{[t_{\eta_k},1]}(d_{\eta_k\ell}^{[t_{\eta_k},1]}) \\ &= a_{\eta_k\ell k} + y_{\eta_k\ell} - \sum_{d_{\eta_k\ell}^{[t_{\eta_k},1]}=0}^{a_{\eta_k\ell k}+y_{\eta_k\ell}-1} F_{\eta_k\ell}^{[t_{\eta_k},1]}(d_{\eta_k\ell}^{[t_{\eta_k},1]}) \end{aligned} \tag{19}$$

where $a_{\eta_k\ell k} + y_{\eta_k\ell}$ is the total available quantity and $\sum_{d_{\eta_k\ell}^{[t_{\eta_k},1]}=0}^{a_{\eta_k\ell k}+y_{\eta_k\ell}-1} F_{\eta_k\ell}^{[t_{\eta_k},1]}(d_{\eta_k\ell}^{[t_{\eta_k},1]})$ is the expected quantity not sold.

Let us now express the inventory level $y_{\eta_k\ell}$ in terms of the initial shipment, i.e. $y_{\eta_k\ell} = (Q^0_{\eta_k\ell} - d_{\eta_k\ell}^{[0,t_{\eta_k}]})^+$ and calculate the expected unconditional sales, $\ddot{V}_{\eta_k\ell}^{[t_{\eta_k},1]}(a_{\eta_k\ell k}, Q^0_{\eta_k\ell})$:

$$\begin{aligned}\ddot{V}_{\eta_k\ell}^{[t_{\eta_k},1]}(a_{\eta_k\ell k}, Q^0_{\eta_k\ell}) &= (1 - F_{\eta_k\ell}^{[0,t_{\eta_k}]}(Q^0_{\eta_k\ell}))\dot{V}_{\eta_k\ell}^{[t_{\eta_k},1]}(a_{\eta_k\ell k}, 0, Q^0_{\eta_k\ell}) + \\ &+ \sum_{d_{\eta_k\ell}^{[t_{\eta_k},1]}=0}^{Q^0_{\eta_k\ell}} f_{\eta_k\ell}^{[0,t_{\eta_k}]}(d_{\eta_k\ell}^{[t_{\eta_k},1]})\dot{V}_{\eta_k\ell}^{[t_{\eta_k},1]}(a_{\eta_k\ell k}, Q^0_{\eta_k\ell} - d_{\eta_k\ell}^{[t_{\eta_k},1]}, Q^0_{\eta_k\ell}) \end{aligned} \tag{20}$$

The first term represents the case where an empty inventory is encountered upon arrival. The second term represents the case when a positive inventory level of $Q^0_{\eta_k\ell} - d_{\eta_k\ell}^{[t_{\eta_k},1]}$ is encountered. For all other retailers on route k, we calculate the expected sales as given in constraints (9), (14) and (15). The total profit can be calculated after solving the recursion by applying (8).

4.2.2 Bounds

The profit calculations are very time consuming. We therefore use the information from the lower and upper bound to investigate if a single move can be profitable in the first place. The lower

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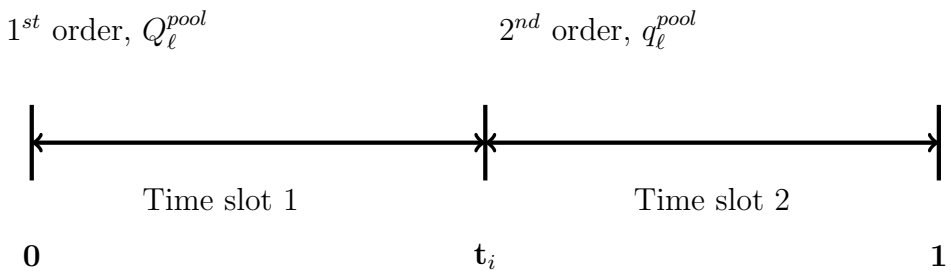


Figure 3: Two ordering opportunities

bound on the profit at a certain retailer is a simple newsvendor solution obtained when only the initial delivery is considered and transshipments are not allowed. We further use an approximation of profit based on a theoretical upper bound where transportation costs are neglected. An upper bound on profit can be obtained by considering both transshipments and inventory pooling. If all retail stores are consolidated into one location and hence use the same inventory, the aggregated uncertainty will be reduced and the overall profit will increase. Let μ_ℓ^{pool} be the mean value and σ_ℓ^{pool} the standard deviation of the pooled demand. They can be calculated as follows:

$$\mu_\ell^{pool} = \sum_{i=1}^{n} \mu_{i\ell} \qquad \forall \ell = 1, .., m \tag{21}$$

$$\sigma_\ell^{pool} = \sqrt{\sum_{i=1}^{n} \sigma_{i\ell}^2} \qquad \forall \ell = 1, .., m \tag{22}$$

Let us now assume that this pooled retailer can wait until some uncertainty is revealed and place an additional shipment during the operation time window (see Figure 3). Hing-Ling Lau and Lau (1998) have proposed a model that captures the decisions how much to order initially, when to order a resupply and how large the size of the resupply should be.

Let t^{opt} be the optimal timing of the second delivery, and Q_ℓ^{pool} be the first and q_ℓ^{pool} the second order quantity for product ℓ. The expected profit $\bar{\Pi}_\ell^{[0,t^{opt}]}(Q_\ell^{pool})$ for the first subperiod for every product ℓ can be calculated as:

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$$\bar{\Pi}_{\ell}^{[0,t^{opt}]}(Q_{\ell}^{pool}) = (p_{\ell} - c_{\ell})Q_{\ell}^{pool} - p_{\ell} \sum_{d_{\ell}^{[0,t^{opt}]}=0}^{Q_{\ell}^{pool}-1} F_{\ell}^{[0,t^{opt}]}(d_{\ell}^{[0,t^{opt}]}) \tag{23}$$

The expected profit in the second subperiod is determined as follows:

$$E_{d_{\ell}^{[0,t^{opt}]}} \bar{\Pi}_{\ell}^{[t^{opt},1]}\Big((Q_{\ell}^{pool} - d_{\ell}^{[0,t^{opt}]})^{+}\Big) = \Big(1 - F_{\ell}^{[0,t^{opt}]}(Q_{\ell}^{pool})\Big)\bar{\Pi}^{[t^{opt},1]}(0) + \\ + \sum_{d_{\ell}^{[0,t^{opt}]}=0}^{Q_{\ell}^{pool}} f_{\ell}^{[0,t^{opt}]}(d_{\ell}^{[0,t^{opt}]})\bar{\Pi}^{[t^{opt},1]}(Q_{\ell}^{pool} - d_{\ell}^{[0,t^{opt}]}) \tag{24}$$

where $\bar{\Pi}_{\ell}^{[t^{opt},1]}(y_{i\ell})$ is calculated as:

$$\bar{\Pi}_{\ell}^{[t^{opt},1]}(y_{\ell}) = \max_{q_{\ell}^{pool} \geq 0} \left\{ -c_{\ell}q_{\ell}^{pool} + p_{\ell}(y_{\ell} + q_{\ell}^{pool}) - p_{\ell} \sum_{d_{\ell}=0}^{y_{\ell}+q_{\ell}^{pool}-1} F_{\ell}^{[t^{opt},1]}(d_{\ell}^{[t^{opt},1]}) \right\}, 0 \leq y_{\ell} \leq Q_{\ell}^{pool} \tag{25}$$

The entire profit achieved in both subperiods is the upper bound for our profit:

$$\Pi_{UB} = \sum_{\ell=1}^{m} \max_{Q_{\ell}^{pool} \geq 0} \left(\bar{\Pi}_{\ell}^{[0,t^{opt}]}(Q_{\ell}^{pool}) + E_{d_{\ell}^{[0,t^{opt}]}} \bar{\Pi}_{\ell}^{[t^{opt},1]}\Big((Q_{\ell}^{pool} - d_{\ell}^{[0,t^{opt}]})^{+}\Big) \right) \tag{26}$$

To reduce the number of possible states, we consider additional bounds on delivery quantities. Let us assume that the first retailer on route k has index ι_k and let $\mathcal{L}_{ik}$ be the set of all retail stores left to be visited on route k **after** store i is visited. The size of the initial shipment to this retailer is bounded by the newsvendor quantities that can be expected during intervals $[0, 1]$ and $[t_i, 1]$:

$$Q_{\iota_k \ell}^{0} \leq q_{\iota_k \ell}^{[0,1]} + \sum_{i \in \mathcal{L}_{\iota_k k}} q_{i\ell}^{[t_i,1]} \qquad \forall k = 1, .., r \tag{27}$$

where $q_{i\ell}^{[t_1,t_2]}$ is the optimal newsvendor quantity in time interval $[t_1, t_2]$ and can be calculated as

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follows:

$$q_{i\ell}^{[t_1,t_2]} = \arg\min_{u\geq 0}\left\{F_{i\ell}^{[t_1,t_2]}(u) \geq \frac{p_\ell - c_\ell}{p_\ell}\right\} \tag{28}$$

The size of the transshipment at the first retailer ι_k is bounded by:

$$0 \geq Q_{\iota_k\ell}^{t_{\iota_k}} \geq -\sum_{i\in\mathcal{L}_{\iota_k k}} q_{i\ell}^{[t_i,1]} \qquad \forall k = 1,..,r \tag{29}$$

The transshipment quantity at the first retailer is negative or zero. In the extreme case of zero remaining inventory at time t_i at each other retailer i, the transshipment should be the optimal quantity that satisfies the demand in period $[t_i, 1]$.

Using the same idea, we can estimate the bounds on quantities for every other retailer i:

$$Q_{i\ell}^{0} \leq q_{i\ell}^{[0,1]} + \sum_{i\in\mathcal{L}_{ik}} q_{i\ell}^{[t_i,1]} \qquad \forall k = 1,..,r \tag{30}$$

Note that the allocation subproblem is solved to optimality whenever a new timing of dispatch is assumed. An additional problem is to find the right timing for the dispatch of vehicle k, t_{0k}. In the majority of vehicle routing problems, the timing of dispatch is simply set to 0. In our problem, the expected revenues depend on the timing of transshipments and, therefore, we have to optimize the timing as well. The optimum can only be found by grid search; however, this is time consuming. In our previous work, we proposed a method for obtaining good solutions based on bisection. Through bisection, we can quickly find a near-optimal time interval for starting the dispatch.

4.3 Shaking Phase

In the **Shaking** phase, we perturb the solution by using one of the following operators: Inter-route **Two-opt**, Inter-route **Swap** or Inter-route **Relocate**. The selection of the operator is done randomly and operators are performed only between two routes.

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The Inter-route **Two-opt** operator deletes one edge from both routes and reconnects the routes so that the direction is preserved (also known as **Two-opt*** operator). The idea of Inter-route **Two-opt** is depicted in Figure 4. The procedure is repeated until we reach the hth neighborhood; no edge can be addressed more than once during the same iteration.



Figure 4: Inter-route Two-opt (Two-opt*)

The Inter-route **Swap** operator selects h retail stores from each route and exchanges their places. No store can be exchanged more than once during an iteration. The procedure is depicted in Figure 5. Both the Inter-route **Two-opt** and the Inter-route **Swap** operators can only be used on solutions that consist of two or more routes. Note that both operators only allow exchanges which are feasible, i.e. for which the maximal route durations are not violated. If no more exchanges are possible, the procedure can stop before reaching the hth neighborhood.



Figure 5: Inter-route Swap

Within the Inter-route **Relocate** operator, up to h retail stores from one route are reinserted into another route (see Figure 6a and b). This operator is also responsible for changing the number of routes. When selecting routes, one of them is allowed to be an empty (additional) route (Figure 6c). Moreover, we can relocate all retailers from one route to another within this operator and, hence, delete a route (Figure 6d). Note that we can relocate retail stores to another route as long as the maximal route duration is not violated.

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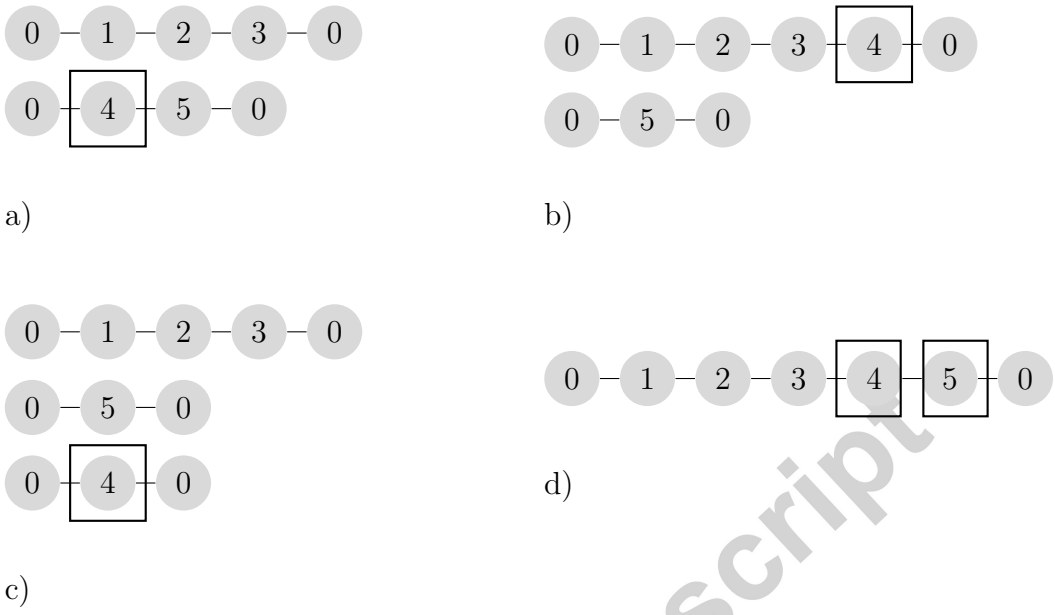


Figure 6: Inter-route Relocate

After the perturbation of the solution, we recalculate the total profit by applying **Allocation** and then search for the local optimum.

4.4 Local Search

In the **Local Search** phase, we randomly select one of the operators: Intra-route **Two-Opt**, Intra-route **Swap** or Intra-route **Relocate**. Due to high computing times, local search operators are only performed within a single route. We only apply the operators to the routes that have been changed in the shaking procedure. The solution is accepted if it is better than the incumbent solution, or if it has never been visited before.

Intra-route **Two-Opt** consists of removing two edges from the tour and reconnecting the tour by inserting two new edges. The procedure is repeated for all pairs of edges in the solution. If an improvement is found, the procedure starts over with the new solution. The ideas of the local search operators are shown in Figures 7 - 9.

Intra-route **Swap** exchanges the positions between two retail stores on the tour. This is done for all pairs of retail stores on the route. Furthermore, within the Intra-route **Relocate** operator,

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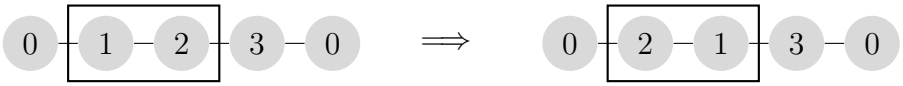


Figure 7: Intra-route Two-opt

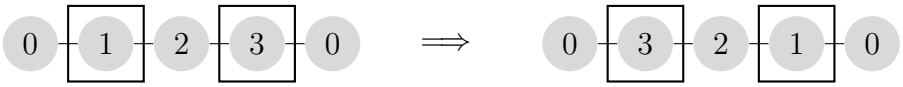


Figure 8: Intra-route Swap

one store is removed from the tour and reinserted at another position. Within every local search operator, we look for the best neighbor solution (see Hansen and Mladenović (2006)).

It is important to discuss the measure of improvement, i.e. the gain obtained from the new solution. As our objective is to maximize the expected total profit and as the profit depends on the sequence in which retail stores are visited, we need to recalculate the profit each time we consider a move in order to calculate the actual gain. As the profit recalculations are very time consuming, we try to extract as much information as possible from our theoretical bounds as calculated in Section 4.2.2. When making an arc interchange, we first define a pool of promising interchanges all of which are feasible solutions with positive gain γ. γ is defined as follows:

$$\gamma = \Pi_{UB} - \Pi(s) + \alpha - \beta \geq 0 \tag{31}$$

where α stands for the costs of vertices that will be deleted, β stands for the costs of vertices that will be inserted, Π_{UB} is the upper bound and $\Pi(s)$ is the profit of the current solution. We select a new solution out of this pool by applying Roulette wheel selection. We favor the solutions with a higher γ. Only after a new solution has been selected do we recalculate the actual profit by applying **Allocation**.

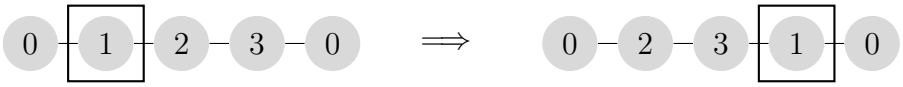


Figure 9: Intra-route Relocate

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4.5 Additional Operators

The operators in Subsections 4.3 and 4.4 are standard in vehicle routing. In order to deal with the profitable tour aspect, additional operators help with the retailer selection. With the **Disturb** operator, we remove retailers from the solution. We propose two **Disturb** operators: **Worst** and **Random**. The **Worst** operator selects retailers based on two criteria: The difference between the upper and the lower bound and the distance from the location to the center of the current tour. Within this operator, a certain percentage of retail stores is deleted from the solution. Additionally, the **Random** operator deletes several entire routes from the solution. The routes are selected by Roulette Wheel selection based on their size, with smaller routes being more likely to get selected.

Within the **Re-insert** operator, additional retailers are added to the solution. We list all the retailers that have not been visited in the ascending order of the transportation costs needed for visiting them and add them to the solution by using the Roulette Wheel selection as long as there is an improvement in total profit. When no more retailers can be inserted, a local search operator is applied and the procedure starts again. We apply this operator only to the routes that have been changed during the shaking procedure. This operator can also be considered as the last shaking operator. When no more stores can be inserted, the entire procedure starts over again until the MAX_LOOP number of iterations has been reached.

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5 Numerical Study

5.1 Parameter Setting

For comparison, we use the same instances as Turan et al. (2015), which are downloadable at "http://prolog.univie.ac.at/links". These are adapted capacitated vehicle routing problem (CVRP) instances, proposed in Christofides (1979). The geographic information about each retail store (x- and y-coordinates) are the same as in the CVRP instances and the mean value of demand is equal to the deterministic demand used in the CVRP instances. Other parameters are selected

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to fit the nature of the problem (see Table 1). The locations contain one warehouse and multiple retail stores. The number of retail stores is set to 5, 10, 15, 20, 30, 40 and 50 and they all sell the same single product. The unit sales price, p, is the same for all settings and set to 10, whereas unit procurement costs, c, vary between high and low. The coefficient of variation, cv, is set to three different values. Furthermore, in order to diversify the transportation costs within different settings, we multiply them with the cost factor, cf, which alternates between five values.

To model the demand, we use a Compound Poisson demand process in which interarrival times are exponentially distributed and transaction sizes are geometrically distributed. This stuttering Poisson process is modeled by using the total demand distribution over a period of length t_i as in Axsäter (2006):

$$P(D(t_i) = v) = \sum_{j=0}^{v} \frac{(\lambda_i t_i)^j}{j!} e^{-\lambda t_i} u_{iv}^j \tag{32}$$

$$u_{iv}^b = \sum_{j=b-1}^{v-1} u_{ij}^{b-1} u_{i,v-j}^1, \qquad b = 2, 3, .., \ v = 1, 2.., \tag{33}$$

where u_{iv}^b denotes the probability that b customers have a total demand of v at store i. The probability that one customer will have a demand of size d, is equal to:

$$u_{id}^1 = g_i(1 - g_i)^{d-1}, \qquad d = 1, 2... \tag{34}$$

where mean transaction size is $\frac{1}{g_i}$.

Furthermore, parameters λ_i and g_i can be calculated as:

$$g_i = \frac{2\mu_i}{\mu_i + \sigma_i^2}, \qquad \lambda_i = \frac{2\mu_i^2}{\mu_i + \sigma_i^2} \tag{35}$$

where μ_i is the mean value and σ_i is the standard deviation of the total period demand at store i.

The parameters in our VNS algorithm are chosen in the following way: Based on the ideas of

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Table 1: Parameters for test instances

Parameters	Values
Number of retail stores	{5, 10, 15, 20, 30, 40, 50}
Unit sales price, p	10
Unit procurement cost, c	{2, 5, 8}
Coefficient of variation, cv	{0.2, 0.5, 1}
Cost factor, cf	{0.1, 0.2, 0.5, 0.8, 1}
MAX_LOOP	20
h^{max}	3

Vansteenwegen et al. (2011) and Vansteenwegen et al. (2009a), we include a *Stopping criterion* of 20 iterations or 30 hours of computations. The stopping criteria of 30 hours is needed for large instances in extreme cases of a demand uncertainty of 1 and low procurement costs of 2. However, as we make all decisions only by using historical data and do not include any information update, we do not need to compute the solution in real time and therefore can afford longer computational times. The maximum neighborhood size, h^{max}, is set to 3. Within the **Disturb** operator, we destroy 20 - 40 % of the solution. $DISTURB_LOOP$ depends on the applied **Disturb** operator: **Worst** or **Random**. The **Worst** operator is applied every third iteration and the **Random** operator every fifth iteration.

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5.2 Computational Results

We compare our results with the results obtained in Turan et al. (2015), where only resupply and no transshipments are considered. Furthermore, we compare our results with the lower bound, which is a simple newsvendor solution obtained when only one initial delivery is considered, and no transshipments nor resupply are allowed.

In Figure 5.2, we investigate the differences in allocation of initial delivery quantities with and without transshipments. We choose a route on which 5 retail stores are visited. In the transshipments case, the first retail store on the route receives a significantly bigger initial delivery than in no transshipments case. The other retail stores on the route get smaller initial quantities

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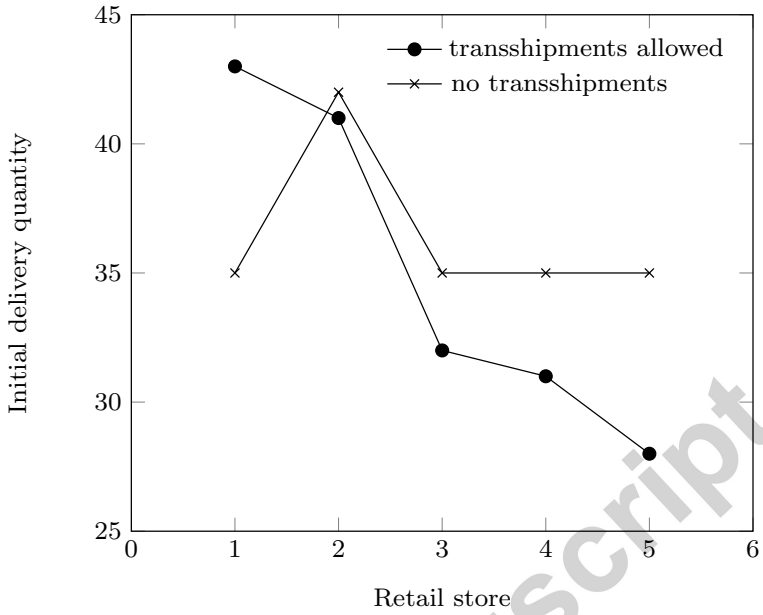


Figure 10: Initial delivery quantity with and without transshipment for each of the five retail stores visited on a given route

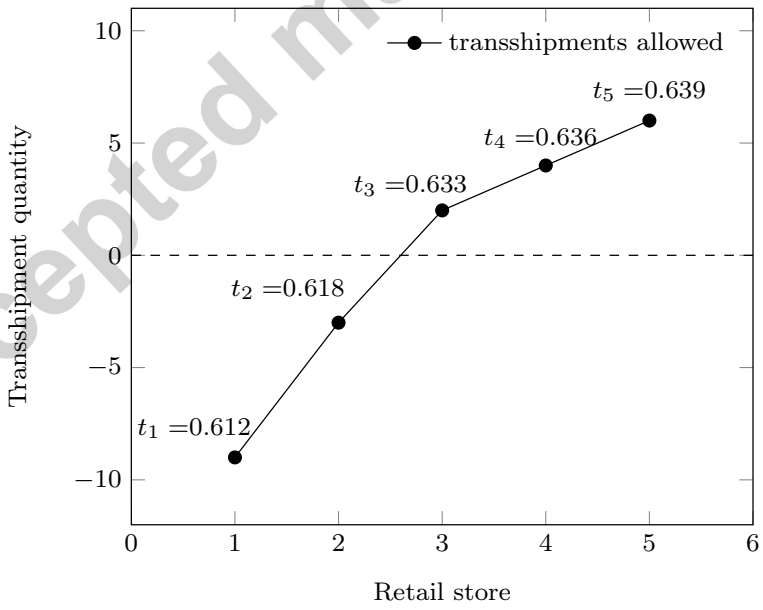


Figure 11: Inventory re-balancing values for each retail store for a given route

than in the simple newsvendor case where no transshipments are allowed. Towards the end of the route, the initial quantities have a tendency to decrease. In sum, the total amount of delivered items is smaller in the case in which transshipments are allowed, as the risk pooling reduces

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Table 2: Comparison to the lower bound and Turan et al. (2015)

n	Gap to LB	Gap to Turan et al. (2015)
5	1.78 %	0.92 %
10	3.22 %	1.18 %
15	3.84 %	1.16 %
20	4.07 %	1.11 %
30	4.25 %	1.09 %
40	4.21 %	1.04 %
50	4.38 %	1.01 %
Avg.	3.55 %	1.07 %

inventories. Note that all retailers have to receive initial delivery until time 0.

The right size of transshipments can only be obtained after we observe the inventory level. In Figure 11, we depict the inventory re-balancing values, calculated at time 0 with the expected inventory level at the time of vehicle arrival. The size of the transshipment is chosen in such a way that the expected stockout at the current retailer and at all other retailers still left to be visited is minimized. It can be observed that the transshipment size is negative at the first two retail stores, as they serve as a "depot" to replenish other stores. The inventory re-balancing is done between $t_1 = 0.612$ and $t_5 = 0.639$.

The aggregated results are given in Tables 2 and 3. Table 2 shows the obtained gap to the lower bound (LB), which is a simple newsvendor solution, and to Turan et al. (2015). In the first column, the number of retail stores is given. The second column shows the gap to the lower bound. This gap is calculated as "(Objective value - LB)/LB". The average improvement over the lower bound is 3.55%. It is evident that the gap to the lower bound increases with the number of retail stores. In the third column, "Gap to Turan et al. (2015)", we show the average improvement compared to the results obtained in Turan et al. (2015), calculated as "(Objective value - Objective value Turan et al. (2015)/Objective value Turan et al. (2015)". We managed to improve the results of the resupply case by 1.07% on average. It can be noticed that this gap is decreasing with a higher number of retail stores. This is due to the fact that the additional benefit in profit is already captured by resupply.

In Table 3, we show details of our results and the results obtained in the resupply case. The average number of visited retail stores increases from 15.48 to 17.04. The average procurement

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Table 3: Comparison to Turan et al. (2015) - aggregated results

	Transshipments				Turan et al. (2015)			
n	expected sales	procurement costs	transport costs	♯ visited retailers	expected sales	procurement costs	transport costs	♯ visited retailers
5	930.03	495.74	11.63	4.52	923.61	493.53	11.27	4.35
10	1854.43	986.97	16.20	8.51	1836.74	980.05	15.38	8.05
15	2825.83	1503.45	18.49	12.43	2800.23	1494.34	16.99	11.46
20	3780.65	2011.75	21.14	16.59	3745.36	1998.03	18.80	14.52
30	5637.45	3000.91	24.24	23.36	5589.23	2982.18	23.01	20.86
40	7524.34	4005.75	32.20	32.55	7461.37	3980.90	29.82	28.79
50	9377.62	4991.58	33.86	38.76	9304.61	4965.13	30.89	35.97
Avg.	3950.79	2102.91	20.97	17.04	3917.08	2090.15	19.43	15.48

costs in the case of transshipments of 2102.91 are slightly higher than the procurement costs in the case of resupply, i.e. 2090.15. These additional costs, however, are compensated by noticeably higher expected sales of 3950.79. It can be concluded that, in the transshipments case, more retail stores are visited and more units are bought, but we also expect that more units will be sold.

We also show a detailed analysis of the obtained improvements, in which we vary the procurement costs, c, the cost factor, cf, and the coefficient of variation, cv. When procurement costs are low (equal to 2), the additional benefit is low in comparison to the resupply case, only 0.56% (see Table 4). For c=5, the additional benefit increases by 1.69% and for c=8, the benefit is even as high as 2.94%.

In Table 5, we show how the improvement changes with different cost factors. For all three cost factors, we multiply the transportation costs with 0.2, 0.5 and 1. The average improvement upon the resupply case is similar, with 1.1% on average. Column "Gap to LB" shows the effect of costs on the solution quality. For a low cost factor of 0.2, the gap to the lower bound is higher, and for a high cost factor of 1, the gap to the lower bound decreases. It is interesting to note, that the benefit compared to the single delivery case (lower bound) is significantly influenced by the cost factor, while the additional benefit compared to the restocking case is almost negligible.

Table 6 shows the effects of the coefficient of variation, i.e. of demand uncertainty, on the solution quality. When cv is small (equal to 0.2), the additional benefit is low, with only 0.38% on average. As demand uncertainty increases, the additional benefit is higher, with 1.52% for cv=0.5 and 1.82% for cv=1.

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Table 4: Detailed analysis - procurement costs, c

	$c = 2$		$c = 5$		$c = 8$	
n	Gap to LB	Gap to Turan et al. (2015)	Gap to LB	Gap to Turan et al. (2015)	Gap to LB	Gap to Turan et al. (2015)
5	0.76 %	0.51 %	1.92 %	1.41 %	2.65 %	2.38 %
10	1.40 %	0.63 %	3.75 %	1.86 %	4.51 %	3.13 %
15	1.73 %	0.62 %	4.40 %	1.78 %	5.41 %	3.24 %
20	1.75 %	0.57 %	4.45 %	1.72 %	6.00 %	3.22 %
30	1.86 %	0.59 %	4.72 %	1.70 %	6.18 %	2.89 %
40	1.83 %	0.52 %	4.77 %	1.67 %	6.05 %	2.84 %
50	1.88 %	0.44 %	5.02 %	1.73 %	6.23 %	2.91 %
Avg.	1.55 %	0.56 %	4.00 %	1.69 %	5.11 %	2.94 %

Table 5: Detailed analysis - cost factor, cf

	$cf = 0.2$		$cf = 0.5$		$cf = 1$	
n	Gap to LB	Gap to Turan et al. (2015)	Gap to LB	Gap to Turan et al. (2015)	Gap to LB	Gap to Turan et al. (2015)
5	3.44 %	0.97 %	1.15 %	0.98 %	0.03 %	0.81 %
10	4.79 %	1.19 %	3.32 %	1.26 %	0.86 %	1.10 %
15	5.09 %	1.20 %	3.90 %	1.25 %	1.97 %	1.03 %
20	5.26 %	1.15 %	4.13 %	1.17 %	2.34 %	1.08 %
30	5.19 %	1.09 %	4.34 %	1.02 %	3.23 %	1.17 %
40	5.01 %	1.00 %	4.39 %	1.04 %	3.25 %	1.06 %
50	5.05 %	1.06 %	4.50 %	1.04 %	3.59 %	0.93 %
Avg.	4.83 %	1.09 %	3.68 %	1.11 %	2.18 %	1.03 %

Table 6: Detailed analysis - coefficient of variation, cv

	$cv = 0.2$		$cv = 0.5$		$cv = 1$	
n	Gap to LB	Gap to Turan et al. (2015)	Gap to LB	Gap to Turan et al. (2015)	Gap to LB	Gap to Turan et al. (2015)
5	1.34 %	0.40 %	2.67 %	1.21 %	1.32 %	1.54 %
10	2.37 %	0.52 %	5.08 %	1.59 %	2.21 %	1.93 %
15	2.90 %	0.51 %	6.02 %	1.58 %	2.61 %	1.86 %
20	2.94 %	0.44 %	6.04 %	1.53 %	3.23 %	1.84 %
30	2.75 %	0.23 %	6.51 %	1.74 %	3.49 %	1.85 %
40	2.88 %	0.23 %	6.54 %	1.59 %	3.22 %	1.84 %
50	3.24 %	0.35 %	6.63 %	1.35 %	3.27 %	1.85 %
Avg.	2.56 %	0.38 %	5.45 %	1.52 %	2.65 %	1.82 %

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6 Conclusions and Future Perspectives

In this paper, we studied a two-echelon inventory routing problem with uncertain demand that consists of one wholesaler and multiple retail stores. We presented a model that helps to better deal with uncertainty by allowing transshipments between retail stores. Each retail store received an initial supply quantity. At a certain point of time during the day, the stores had the option to re-balance their inventories by receiving additional items or by reducing unnecessary inventory. The selection of the stores and the sequence in which they are visited had to be decided upfront. Moreover, we determined the number of vehicles needed to execute the transshipments and the timing of the dispatch. Furthermore, we determined the size of the initial supply and the transshipment strategy.

We proposed a mathematical model that captures the decisions to be made, with the goal of maximizing the entire net profit after deduction of procurement and transportation costs. Due to the high complexity of the model, we solved the problem by variable neighborhood search (VNS). In order to keep track of the maximal benefit, we included an operator based on dynamic programming in our VNS procedure, which optimized the expected profit for a given retailer sequence. We further provided a theoretical lower and upper bound, both of which helped to speed up the solution method.

We tested our algorithm on the test instances provided by Turan et al. (2015). They provided 225 randomly generated instances originally based on the instances for the capacitated vehicle routing problem (CVRP). In their paper, they considered a resupply during the operation time window on top of regular supply for a selected pool of retail stores. They also solved both the routing and inventory allocation problems, but allowed no transshipments between retail stores. Our results can be summarized as follows:

- The results showed a noticeable improvement over the results of Turan et al. (2015), with an additional benefit of 1.07% on average.

- The positive impact of transshipments can be seen especially in cases of high demand un-

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certainty and high procurement costs with improvements increasing up to 3%.

- On average, more retail stores are visited than in the resupply case.
- More units are purchased and the expected sales are higher, i.e. the provided service level is larger.
- A structurally different allocation policy is observed: While in the resupply case the initial quantity increases in the same way as the expected demand, now the first retailers on the route receive remarkably higher quantities than the expected demand.

Some ideas for future work include the improvement of inventory information. Due to the technological restrictions, we only worked with upfront information. It might be interesting to investigate the costs of having updated information about actual inventory levels. Furthermore, when having updated information, it could be investigated if an additional benefit could be obtained through considering a re-routing option for the rest of the route. Finally, in cases where all retail stores on one route can be visited within a short amount of time (if retailers are located very close to each other and/or only few retailers have to be visited on one route) we could also consider several transshipment periods along the operation time window of the retailers.

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A NOTATION

Table 7: NOTATION

Abbreviation	Description
Indices	
i, j	Index of a store,$i, j = 0, .., n$; with $i = 0$ depot is denoted
ℓ	Index of a product, $\ell = 1, .., m$
k	Index of a vehicle, $k = 1, .., r$
Data	
$\mu_{i\ell}$	Mean value of demand of product ℓ in store i
$\sigma_{i\ell}$	Standard deviation of demand of product ℓ in store i
p_ℓ	Unit sales price of product ℓ
c_ℓ	Unit procurement costs of product ℓ
cv	Coefficient of variation
cf	Cost factor
tt_{ij}	Travel time between locations i and j
tc_{ij}	Travel costs between locations i and j
Decision variables	
$Q^0_{i\ell}$	Initial shipment for store i
$Q^{t_i}_{i\ell}$	Size of transshipment at time t_i for store i
$y^{t_i}_{i\ell}$	Inventory level at store i at time t_i
$a_{i\ell k}$	Remaining stock of product ℓ on vehicle k upon arrival at store i
t_i	Timing of transshipment for store i
t_{0k}	Timing of dispatch for vehicle k
x_{ijk}	Binary flow variable

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